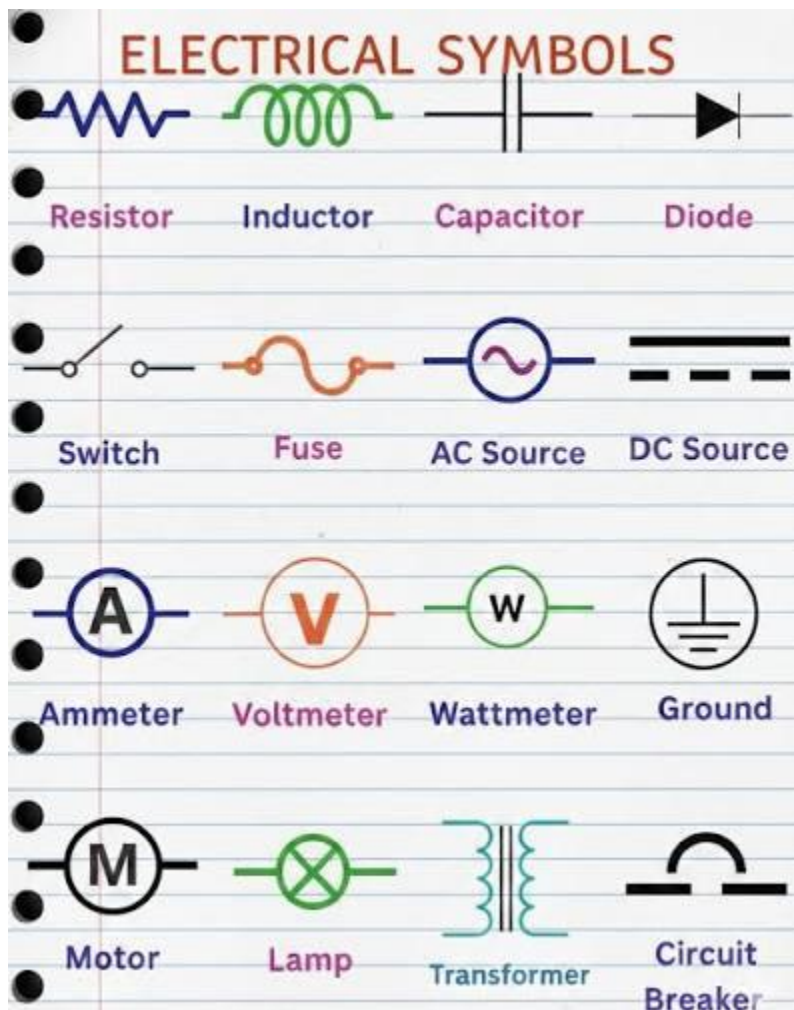


Part I. Basic Electrical Quantities (1–10)

1. Define the following:
 - a. Electric charge
 - Electric charge relates to electric current and time. It is a basic property of matter responsible for electric force. Its conservation charge is neither created nor destroyed.
 - b. Electric current
 - Electric current in a wire, where the charge carriers are electrons, measures the quantity of the charge that passes any point of the wire per unit of time it takes.
 - c. Voltage
 - The amount of voltage is measured by the unit of volt(V), higher voltages can cause more electricity to an electronic device. On the other hand if the voltage is too low, it can cause issues making the devices built around them ineffective.
 - d. Resistance
 - A force that counteracts the flow of the current when an electron differential exists in two terminals, its flow will go from high to low. Resistance may be determined by the type and temperature where the electricity is flowing, as well as its length.
2. State Ohm's Law.
 - Ohm's law states that the voltage across a conductor is **directly proportional** to the current flowing through it, provided all physical conditions and temperature **remain constant**.
3. State Kirchhoff's Current Law (KCL).
 - Kirchhoff's Current law states that the sum of all electric currents meeting at any electrical junction or node in a circuit is equal to exactly 0 .
4. State Kirchhoff's Voltage Law (KVL).
 - Kirchhoff's Voltage law states that the algebraic sum of all voltage charges around any closed loop in an electric circuit is exactly 0.
5. Differentiate an ideal voltage source from an ideal current source.
 - The voltage source and current source are both electrical sources that can provide electrical energy to drive or move an electrical load. An ideal voltage source has two terminals that remains constant and does not depend on the amount of current flowing through the circuit. An Ideal current source provides constant independent voltage across or change in the connected load. It also has infinite internal resistance.
6. What is electrical power?
 - This is defined as the rate where electric energy is transferred through a circuit, it uses power in watts, voltage and current.

7. What is electrical energy? State its SI unit.
- SI unit is Watt, joule per second. Electric power is supplied by electric batteries and produced by electric generators.
8. Explain the difference between conductors, semiconductors, and insulators.
- Conductors let the electricity flow freely, semiconductors can be precisely controlled by temperature, voltage, or added impurities to control the flow of electricity, and insulators can completely block electrical flow.
9. Enumerate at least five passive electrical components.
- Resistors, Capacitors, Oscillators, Microprocessors and Inductors.
10. Draw the schematic symbols of the following:
- Resistor
 - Capacitor
 - Inductor
 - Battery
 - Ground



Part II. Circuit Elements (11–20)

11. Differentiate active elements from passive elements.

- Active elements supply energy or power to a circuit in the form of a voltage or current. Passive elements act as energy acceptors, they are quiet enablers.

12. Define resistance and state the factors affecting its value.

- Resistance is the opposition to the flow of electrons as they move through a material. There are four factors affecting resistance, they are cross-sectional area, length, temperature and the type of the material. As the length increases its resistance also increases. The cross sectional area varies inversely with the material, if the cross sectional area is doubled the resistance is halved. Resistance depends on the type of material used since resistance is equal to resistivity, resistivity is combined above formulas and also measured between two opposite faces of a 1m cube material like copper.

13. State the color bands of a $470\ \Omega \pm 5\%$ resistor.

- The color bands are yellow, violet, brown and gold, yellow is 4, violet is 7, brown is $\times 10$ multiplier and gold positive or negative 5% tolerance.

14. Explain the function of a capacitor in an electrical circuit.

- A capacitor is an electronic component used for storing electrical energy, if positive charges $+Q$ with total charge conductors and an equal amount of negative charge $-Q$ conductor, the capacitor is said to have a charge Q .

15. Explain the function of an inductor in an electrical circuit.

- An inductor is a passive electrical component that can store energy in a magnetic field when the electric flows through it. Its principle states that a changing current creates an opposing voltage that stabilizes the flow, called inductance.

16. Differentiate DC circuits from AC circuits.

- The main difference between DC and AC lies in the direction where the electrons flow. In alternating current, the electric charge flow changes direction periodically. Unlike direct current, the current electricity flows in a single direction in a steady voltage.

17. Draw a simple closed-loop electrical circuit consisting of a battery, switch, and resistor.

- A simple closed-loop electrical circuit consists of a battery acting as the voltage source, a switch to open or close the path, and a resistor to limit current, all connected in a series loop with conducting wires.

18. What happens to current when resistance increases while voltage remains constant?

- As defined by Ohm's law, the resistance increases while voltage stays the same, the current decreases. The current and resistance are inversely proportional, increasing the resistance causes the current to drop.

19. Explain the relationship among voltage, current, and resistance.
- Ohm's law states that current is directly proportional to voltage and inversely proportional to resistance.
20. State three practical applications of electrical circuits in everyday life.
- Home appliances like light systems, refrigerators and freezers, also smartphones.
-

Part III. Circuit Analysis (21–35)

21. Differentiate series circuits from parallel circuits.
- A series circuit is defined as there is only one path for the current to flow, they have the same current flowing throughout all the components.
 - A parallel circuit, there are never more than two sets of electrically common points, no matter how many components are connected. There are many paths for the current flow but only one voltage across all components.
22. State the characteristics of a series circuit.
- It is a simple configuration because it has only one path for current flow. The current flowing in each component has the same value and the voltage in each component in each component depends on the value of the resistor. Though, the main disadvantage of a series circuit is that if one of the electrical loads is disconnected then the electricity flow in the entire circuit will stop.
23. State the characteristics of a parallel circuit.
- A parallel circuit features multiple branches, providing several paths for electric current to flow
24. Compute the equivalent resistance of three resistors connected in series:
- $R_1 = 5\ \Omega$
 - $R_2 = 8\ \Omega$
 - $R_3 = 12\ \Omega$
 - **25Ω**
25. Compute the equivalent resistance of the following parallel resistors:
- $6\ \Omega$
 - $12\ \Omega$
 - **4Ω**
26. A 24-V source supplies a 6- Ω resistor. Determine the current.
- $24 / 6 = \mathbf{4\Omega}$
27. A resistor carries 4 A with a voltage drop of 80 V. Determine its resistance.
- $80 / 4 = \mathbf{20\Omega}$

28. Determine the power consumed by a 120-V load drawing 8 A.

- $120 * 8 = \mathbf{960\ W}$

29. A 100-W lamp operates for 6 hours. Determine the electrical energy consumed in kWh.

- $100 * 6 = 600 / 1000 = \mathbf{0.6kWh}$

30. Using KCL, explain what happens to current entering and leaving a node.

- Based on the conservation of electric charge, the total current entering a node must exactly equal the total current leaving the node. Charge cannot accumulate at a single point in a circuit.

31. Using KVL, explain why the algebraic sum of voltages around a closed loop is zero.

- As the electric charge travels around any closed loop and returns to its starting point, it gains energy from voltage sources and loses energy across voltage drops.

32. Differentiate an open circuit from a short circuit.

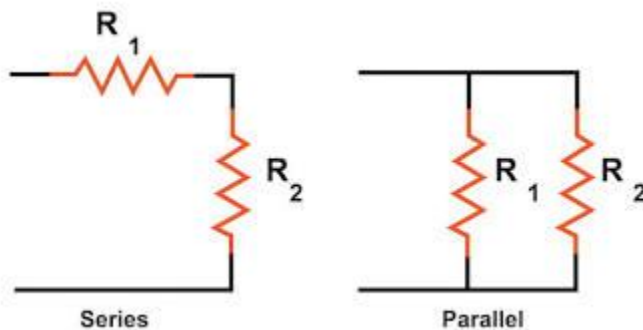
- An open circuit has a gap in the electrical path. It has infinite resistance that prevents any current from flowing even if a voltage is present.
- A short circuit has near zero resistance that causes a massive and uncontrolled rush of electrical current.

33. Explain why a short circuit is dangerous.

- Because of its high current, it may cause an instant high level spike, its extreme heat, according to Joule's law, high current generates intense thermal energy in wires and components and this heat can melt wire insulation that can permanently destroy electrical equipment.

34. Draw one example each of:

- **Series circuit**
- **Parallel circuit**



35. State one practical application of a series circuit and one practical application of a parallel circuit.

- String lights; Series circuit

- Household wiring; Parallel circuit
-

Part IV. Network Theorems and Measuring Instruments (36–45)

36. State the purpose of Thevenin's Theorem.

- It simplifies any complex linear circuit into a single voltage source in series with a single resistor.

37. State the purpose of Norton's Theorem.

- Simplifies any complex linear circuit into a single ideal current source in parallel with a single resistor.

38. Differentiate Thevenin's equivalent circuit from Norton's equivalent circuit.

- Thevenin's Circuit uses a voltage source in series with an equivalent resistance.
- Norton's Circuit uses a current source in parallel with an equivalent resistance.

39. Differentiate an ideal ammeter from an ideal voltmeter.

- Ideal Ammeter: Has zero internal resistance so it does not restrict current flow through the branch it measures.
- Ideal Voltmeter: Has infinite internal resistance so it draws zero current away from the circuit branch it measures.

40. Explain the proper way of connecting:

- Ammeter - Must be connected in series with the load so the entire branch current passes directly through it.
- Voltmeter - Must be connected in parallel across the component to measure the potential difference between two specific points.

41. Define continuity testing.

- A quick check to verify if a complete, unbroken electrical path exists between two points in a circuit.

42. What instrument is commonly used to measure:

- Voltage - Voltmeter
- Current - Ammeter
- Resistance - Ohmmeter

43. Explain the importance of calibration in electrical measuring instruments.

- Adjusts the instrument to match a certified reference standard, reducing measurement errors.
- Ensures readings are dependable over time, preventing false passes, hazardous overloads, or incorrect diagnostics.

44. State three safety precautions when performing circuit measurements.

- Power Check: De-energize the circuit completely before connecting an ammeter or ohmmeter to avoid arcing or blowing internal fuses.
- Proper Range Setting: Set the meter to its highest measurement range first if the expected voltage or current level is unknown.
- PPE Usage: Wear protective gear, use insulated test leads, and handle probes with one hand when dealing with live, high-voltage systems.

45. Why should measuring instruments be connected properly before energizing a circuit?

- Misconnecting a low-resistance ammeter in parallel acts like a short circuit, destroying the meter or tripping a breaker. Additionally, severe short circuits can cause unexpected electrical flashes, device explosions, or user shocks.

Part V. Problem Solving and Critical Thinking (46–50)

46. Three resistors of $10\ \Omega$, $15\ \Omega$, and $20\ \Omega$ are connected in series to a 90-V source.

Determine:

- Total resistance: $10 + 15 + 20 = \mathbf{45\ \Omega}$
- Circuit current: $90 / 45 = \mathbf{2\ A}$
- Voltage drop across each resistor: $2 * 10 = \mathbf{20\ V}$, $2 * 30 = \mathbf{60\ V}$, $2 * 20 = \mathbf{40\ V}$

47. Two resistors, $8\ \Omega$ and $12\ \Omega$, are connected in parallel across a 24-V source.

Determine:

- Equivalent resistance: $96 / 20 = \mathbf{4.8\ \Omega}$
- Total current supplied by the source: $24 / 4.8 = \mathbf{5A}$

48. A circuit consumes 720 W while operating at 240 V.

Determine the current drawn by the circuit.

- $720 / 240 = \mathbf{3\ A}$

49. A resistor has a resistance of $50\ \Omega$ and carries 2.5 A.

Determine:

- Voltage across the resistor: $2.5 * 50 = \mathbf{125\ V}$
- Power dissipated: $2.5^2 * 50 = \mathbf{312.5\ W}$

50. In your own words, explain why a strong foundation in Fundamentals of Electrical Circuits is essential before studying advanced Electrical Engineering courses such as Electrical Machines, Power Systems, Electronics, and Transmission and Distribution System Design.

A strong foundation in EE circuits can help us to be good critical thinkers in times of action, for example, creating a machine, a system, electronics, and etc. With the use of our solid foundation we can become mathematically and conceptually very good to navigate through our way out the problem we will encounter in the future. For this we can analyze the behaviour of the electrical circuits, we can vividly analyze the calculations of the voltage and currents. As we deal with power systems, we could operate a three phase AC by understanding phase angles,

real power and etc, this requires a command of fundamental AC circuit theories and phasor math. As for electronics, utilizing semiconductors devices to process signals and convert power, we would know what to put, to use and to explore how an audio signal moves through an amplifier or how a filter cleans up noise. For me, having a solid foundation through basic information is very important, it will bring to many useful ways to solve a problem.